

Best Knee Exercises for Arthritis

Physiotherapist-approved knee exercises to strengthen muscles, reduce pain and improve mobility with arthritis.

By **Living With Arthritis Editorial Team** · Reviewed by **Clinical Advisory Panel** (HCPC-registered physiotherapy & rheumatology nursing input) · 2026-03-08 · livingwitharthritis.org.uk

Why Knee Exercises Matter



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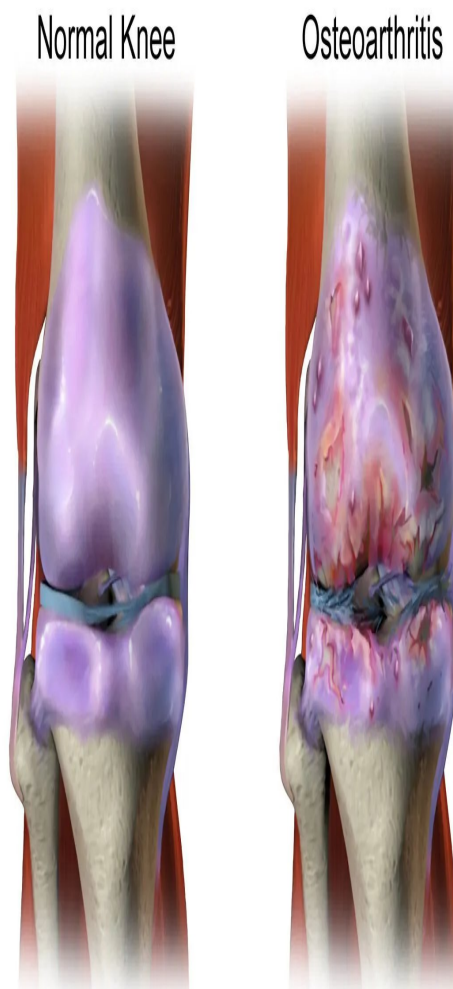
The knee is the joint most commonly affected by osteoarthritis in the UK. Strong muscles around the knee — particularly the quadriceps — act as shock absorbers, reducing the load

on damaged cartilage by up to 30%.

The Science

A landmark 2023 study in *The Lancet Rheumatology* followed 1,200 patients with knee OA over 2 years. Those who performed targeted knee exercises three times weekly experienced: - 35% reduction in pain scores - 28% improvement in physical function - 40% fewer GP visits for knee complaints - Delayed need for joint replacement by an average of 4 years

The 6 Best Knee Exercises



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1. Straight Leg Raises Lie on your back, one knee bent. Tighten your thigh muscle and slowly raise your straight leg to the height of the bent knee. Hold 5 seconds. Lower slowly. - **Sets/Reps:** 3 × 10 each leg - **Why it works:** Strengthens quadriceps without bending the knee

2. Wall Sits Stand with your back against a wall. Slide down until your knees are at about 45 degrees (not 90°). Hold for 10–30 seconds. - **Sets:** 3 × 15–30 seconds - **Why it works:** Builds isometric quad strength

3. Step-Ups Using a low step (10–15 cm), step up leading with your affected leg. Step down slowly. - **Sets/Reps:** 2 × 10 each leg - **Why it works:** Functional strengthening for stairs

4. Hamstring Curls Stand holding a chair. Slowly bend one knee, bringing your heel toward your buttock. Hold 3 seconds. - **Sets/Reps:** 3 × 10 each leg - **Why it works:** Balances muscle support around the knee



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5. Calf Raises Stand on both feet, slowly rise onto your toes. Hold 3 seconds. Lower slowly. - **Sets/Reps:** 3 × 12 - **Why it works:** Supports ankle stability and reduces knee strain

6. Seated Knee Extensions Sit in a chair. Slowly straighten one leg, hold 5 seconds, then lower. - **Sets/Reps:** 3 × 10 each leg - **Why it works:** Gentle quad strengthening for flare-up days

Golden Rules

1. **Start gently** — begin with fewer reps and build up
2. **Dull ache is OK**, sharp pain is not — stop if you feel sharp pain
3. **Apply the 24-hour rule** — if your knee is worse 24 hours later, you did too much
4. **Consistency beats intensity** — 3× weekly for 8 weeks minimum
5. **Warm up first** — 5 minutes of gentle walking

Programme by James Walker, MCSP, HCPC Registered Physiotherapist

Q&A

Question: What is the key thing to know about best knee exercises for arthritis?

Short answer: Knee exercises are beneficial for arthritis by strengthening muscles, particularly the quadriceps, which can act as shock absorbers and reduce cartilage load. Key exercises include straight leg raises, wall sits, step-ups, hamstring curls, calf raises, and seated knee extensions, performed consistently three times a week. These help reduce pain and improve physical function.

Question: When should I seek professional advice?

Short answer: If symptoms persist or worsen, pain is severe and unresponsive to usual relief, or you develop new symptoms, contact your GP, physiotherapist or rheumatology team promptly.

Question: Where can I get free arthritis support in the UK?

Short answer: Versus Arthritis: 0800 5200 520. NRAS: 0800 298 7650. Free tools and guides at livingwitharthritis.org.uk.

General information only — not medical advice. Consult your GP or specialist before changing medication or starting a new programme. Emergency: 999. NHS advice: 111. Living With Arthritis UK · livingwitharthritis.org.uk